Fixed Income Analysis Report

Implicit Differentiation and First-Order Risk

Calibration Jacobians, Volatility Parameters, Discount-Factor Gradients and Cashflow Valuation

This report solves the three written questions in Assignment 5 and documents the completed valuation-engine implementation. It connects Newton calibration to the implicit-function theorem, applies the framework to a three-parameter implied-volatility smile, and verifies analytic curve-node risk for a fixed accrued cashflow on a referenced instantaneous-forward-rate model.

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Abstract

Calibration is a root-finding problem, while calibration risk is an implicit differentiation problem. A nonsingular pricing Jacobian maps market-price changes into model-parameter changes. For a volatility smile, this Jacobian is the product of option vega and the interpolator's parameter gradient. In the programming component, discount factors are exponentials of integrated instantaneous forward rates. Their node gradients are propagated through a referenced funding curve with the product rule and then scaled by the cashflow amount to obtain exact first-order PV risk.

Keywords: Implicit function • Newton calibration • Volatility smile • IFR curve • First-order risk

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1 Q1: Solving the Calibration Equations

Question focus. *Give a general recipe for recovering the implied parameter vector X^I from $V(X^I) = X^M$.*

Define the residual

$$R(X^I) = V(X^I) - X^M. \tag{1.1}$$

Calibration seeks a root, $R(X^I) = 0$. Starting from a reasonable guess X_0^I, calculate the model Jacobian

$$J_V(X_n^I) = \left.\frac{\partial V}{\partial X^I}\right|_{X_n^I} \tag{1.2}$$

and solve the Newton system

$$J_V(X_n^I)\,\Delta X_n^I = X^M - V(X_n^I). \tag{1.3}$$

Update $X_{n+1}^I = X_n^I + \Delta X_n^I$ and repeat until both the pricing residual and the parameter step satisfy the chosen tolerances. A damped step or line search can be used when a full Newton step is unstable. Numerically, the linear system should be solved directly rather than by forming an explicit matrix inverse. Bounds or a reparameterization may be added when admissibility, such as positive volatility, must be enforced.

Local uniqueness and a stable Newton step require a square, nonsingular Jacobian near the solution. A poorly conditioned Jacobian signals that the market observations do not identify the model parameters reliably.

Key result. Calibration consists of defining price residuals, evaluating the pricing Jacobian, solving a linear Newton system, updating the parameters and checking convergence.

2 Q2: Sensitivity of Implied Parameters

Question focus. *Calculate dX^I/dX^M at the calibrated optimum.*

At the optimum, the calibration identity is

$$V(X^I(X^M)) = X^M. \tag{2.1}$$

Taking total differentials gives

$$J_V\, dX^I = dX^M. \tag{2.2}$$

Provided J_V is nonsingular, the implicit-function theorem yields

$$\boxed{\frac{dX^I}{dX^M} = J_V^{-1}.} \tag{2.3}$$

Each column gives the response of all implied parameters to a unit change in one market quote. In implementation, a particular shock is obtained by solving $J_V dX^I = dX^M$ rather than explicitly inverting J_V.

For a downstream quantity $G(X^I)$, the chain rule gives market-quote risk:

$$\boxed{\frac{dG}{dX^M} = \frac{\partial G}{\partial X^I} J_V^{-1}.} \tag{2.4}$$

3 Q3: Three-Point Implied-Volatility Calibration

3.1 Calibration from option prices

Let $B_i(\sigma)$ denote the undiscounted European option pricing function at strike K_i. The three parameters $\theta \in \mathbb{R}^3$ are determined by

$$B_i(\sigma(K_i;\theta)) = P_i, \qquad i = 1,2,3. \tag{3.1}$$

Equivalently, solve the three residual equations

$$R_i(\theta) = B_i(\sigma(K_i;\theta)) - P_i = 0 \tag{3.2}$$

with the Newton procedure from Q1. One may also invert each option price to an implied volatility first and then fit the three interpolator parameters; the direct price-space formulation makes the market-price sensitivity immediate.

The calibration Jacobian is

$$J_{ij} = \frac{\partial P_i}{\partial \theta_j} = \underbrace{\frac{\partial B_i}{\partial \sigma_i}}_{\text{Vega}_i} \frac{\partial \sigma(K_i;\theta)}{\partial \theta_j}. \tag{3.3}$$

Therefore

$$\frac{d\theta}{dP} = J^{-1}, \qquad \boxed{\frac{d\sigma(K)}{dP} = \nabla_\theta \sigma(K;\theta)^\mathsf{T} J^{-1}.} \tag{3.4}$$

The direct sensitivity of the interpolated volatility to its parameters is simply $d\sigma(K) = \nabla_\theta \sigma(K;\theta)^\mathsf{T} d\theta$.

3.2 Trader-interpretable parameters

A natural three-parameter representation is level, skew and curvature:

$$\theta_{\text{ATM}} = \sigma(K_{\text{ATM}}), \tag{3.5}$$

$$\theta_{\text{RR}} = \sigma(K_3) - \sigma(K_1), \tag{3.6}$$

$$\theta_{\text{BF}} = \frac{\sigma(K_1) + \sigma(K_3)}{2} - \sigma(K_{\text{ATM}}). \tag{3.7}$$

The first parameter moves the smile vertically, the risk reversal tilts the two wings relative to one another, and the butterfly changes their average height relative to ATM. This basis makes sensitivities easier to discuss than sensitivities to three opaque polynomial coefficients.

4 Completed Valuation Engine

4.1 Discount-factor gradient

For one instantaneous-forward-rate component,

$$D(T) = \exp\left(-\int_0^T f(u)\,du\right). \tag{4.1}$$

If $w_i(T)$ is the derivative of the integrated interpolant with respect to node ordinate f_i, then

$$\boxed{\frac{\partial D(T)}{\partial f_i} = -D(T)w_i(T).} \tag{4.2}$$

For the referenced funding curve $D_{\text{fund}} = D_{\text{spread}}D_{\text{SOFR}}$, the implementation walks the reference chain and applies the product rule. Each component block receives its local derivative multiplied by every other local discount factor. The supplied `scaler` multiplies only the new contribution, while `accumulate` controls replacement or addition.

4.2 Fixed accrued cashflow

For signed notional N and accrual fraction α, the amount is $C = N\alpha$. A future payment has

$$PV = CD(T_p), \qquad \nabla PV = C\nabla D(T_p). \tag{4.3}$$

A payment today has $PV = Cash = C$ and zero curve risk. A past payment has zero PV, cash and risk.

Table 1. Verified Assignment 5 outputs

Quantity	Result
Cashflow PV	$245{,}870.910580$
Cash today	0
SOFR raw gradient	$(-60{,}625.703979, -123{,}272.264756, 0)$
Funding-spread raw gradient	$(-121{,}925.026890, -61{,}972.941845)$
One-basis-point report	Raw gradients multiplied by 10^{-4}, preserving sign

The zero in the third SOFR node is expected because its piecewise-constant segment begins after the payment date.

5 Conclusion

The same Jacobian that drives Newton calibration also maps market shocks into implied-parameter shocks. In the option example, vega connects price space to volatility space and a level–skew–curvature parameterization makes the resulting risk intuitive. The completed library applies the same derivative discipline to yield curves: differentiate the integrated IFR, propagate the gradient through referenced discount factors, and scale it by the contractual cashflow. The executed notebook confirms the PV and every reported risk block.